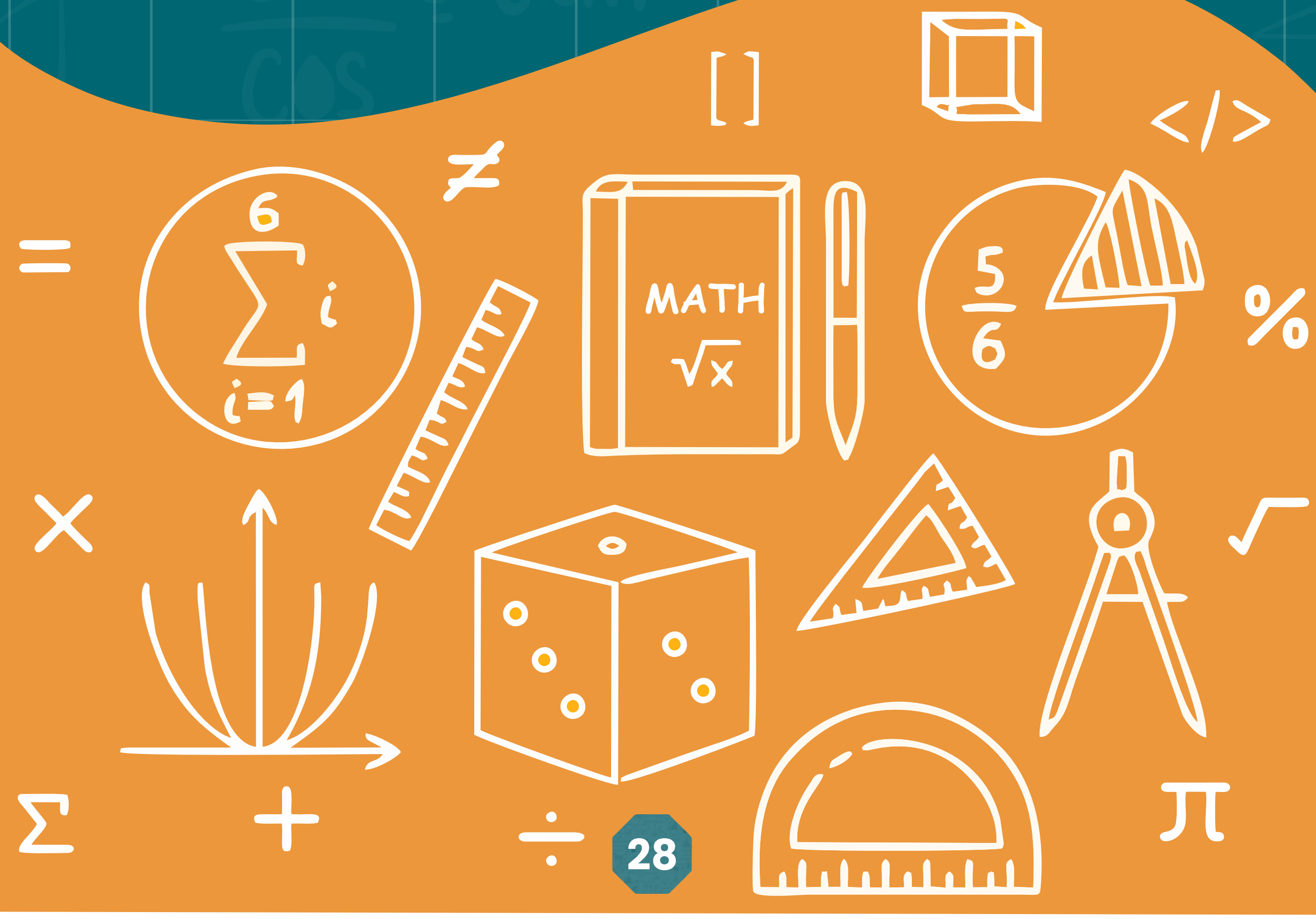
UNIT
3

DECIMAL



CONVERSION OF DECIMALS TO RATIONAL NUMBERS

A rational number is any number that can be expressed as a fraction $\frac{1}{2}$, where p and q are integers, and q≠0. Decimals can be converted into rational numbers by writing them as fractions and simplifying if necessary.

There are two main types of decimals to consider:
1. Terminating decimals (e.g., 0.5, 2.750).
2. Repeating decimals (e.g., 0.333.., 0.909090..).

Example: Convert a decimal 0.5 to a rational number.

Solution: $0.5 = \frac{05}{10} = \frac{5}{2 \times 5} = \frac{\cancel{5}^{1}}{2 \times \cancel{5}^{1}} = \frac{1}{2}$

or $0.5 = \frac{\cancel{5}^{1}}{\cancel{10}^{2}} = \frac{1}{2}$

Example: Convert decimal 2.40 into a rational number.

Solution: Given decimal 2.40 comprises of two parts: '2' (a whole part) and '.40' (a decimal part).

So, we first convert the decimal part '.40' into a rational number.

$40 = \frac{40}{100}$(As .40 has two places of decimal, so we put two zeros after '1').

So, $2.40 = 2 + .40 = 2\frac{40}{100} = 2\frac{\cancel{4}^{2}}{\cancel{10}^{5}} = 2\frac{2}{5} = \frac{12}{5}$

Thus, decimal $2.40 = \frac{12}{5}$, which is a common fraction or a rational number.

Example: Convert -1.60 into a Rational Number.
Here whole part '-1' is kept as a whole and the decimal part '.60' is to be converted into a common fraction.

So, $-1.60 = -(1 + .60)$

$$= -\left(1 + \frac{6\cancel{0}}{10\cancel{0}}\right) = -\left(1 + \frac{\cancel{6}^{3}}{\cancel{10}^{5}}\right)$$

$$= -\left(1 + \frac{3}{5}\right) = -1\frac{3}{5} = -\frac{8}{5}$$

Thus decimal $-1.60 = -\frac{8}{5}$, which is a rational number.

EXERCISE 3.1

Q1. Convert the following one place-decimals into rational numbers:

(1) 0.2 **(2)** 0.4 **(3)** 0.6 **(4)** 0.8 **(5)** 0.9

Q2. Convert the following two place-decimals into rational numbers:

(1) 0.15 **(2)** 0.35 **(3)** 0.48 **(4)** 0.75 **(5)** 0.95

(6) -1.30 **(7)** -2.50 **(8)** 5.05 **(9)** 5.60 **(10)** 6.85

Decimal Fraction

- **Non-terminating** (1.454545..., 3.987..., etc.)
  - **Recurring** (1.4545..., 3.888..., etc.)
  - **non-recurring** (1.4568..., 3.9876..., etc.)
- **Terminating** (1.2, 2.5, 3.9, etc.)

Terminating decimals:

Let us consider the conversion of some rational numbers:

$$\frac{1}{4}, \frac{1}{5}, \frac{1}{2} \text{ and } \frac{3}{8}$$

Conversion of Rational Numbers into Terminating decimal.

(i) $\frac{1}{4} = 1 \div 4$

$$\begin{array}{r} 0.25 \\ 4\overline{)\;10} \\ -8 \\ \hline 20 \\ -20 \\ \hline 0 \end{array}$$

Thus, $\frac{1}{4} = 0.25$

(ii) $\frac{1}{5} = 1 \div 5$

$$\begin{array}{r} 0.2 \\ 5\overline{)\;10} \\ -10 \\ \hline 0 \end{array}$$

Thus, $\frac{1}{5} = 0.2$

(iii) $\frac{1}{2} = 1 \div 2$

$$\begin{array}{r} 0.5 \\ 2\overline{)\;10} \\ -10 \\ \hline 0 \end{array}$$

Thus, $\frac{1}{2} = 0.5$

(iv) $\frac{3}{8} = 3 \div 8$

$$\begin{array}{r} 0.375 \\ 8\overline{)\;30} \\ -24 \\ \hline 60 \\ -56 \\ \hline 40 \\ -40 \\ \hline 0 \end{array}$$

Thus, $\frac{3}{8} = 0.375$

Non-terminating decimals:

Consider the conversion of rational numbers.

$$\frac{7}{9}, \frac{1}{6} \text{ and } \frac{10}{11}$$

Conversion of Rational Numbers into Non-Terminating decimal.

(i) $\frac{7}{9} = 7 \div 9$

$$\begin{array}{r|l} & 0.777... \\ 9 & 70 \\ & -63 \\ \hline & 70 \\ & -63 \\ \hline & 70 \\ & -63 \\ \hline & 7 \end{array}$$

Thus, $\frac{7}{9} = 0.777...$

(ii) $\frac{1}{6} = 1 \div 6$

$$\begin{array}{r|l} & 0.1666... \\ 6 & 10 \\ & -6 \\ \hline & 40 \\ & -36 \\ \hline & 40 \\ & -36 \\ \hline & 4 \end{array}$$

Thus, $\frac{1}{6} = 0.1666...$

(iii) $\frac{10}{11} = 10 \div 11$

$$\begin{array}{r|l} & 0.9090... \\ 11 & 100 \\ & -99 \\ \hline & 100 \\ & -99 \\ \hline & 100 \\ & -99 \\ \hline & 1 \end{array}$$

Thus, $\frac{10}{11} = 0.9090...$

By converting the given rational numbers to decimals, it is found that:

(i) $\frac{7}{9} = 0.777...$ **(ii)** $\frac{1}{6} = 0.1666...$ **(iii)** $\frac{10}{11} = 0.9090...$

Activity: Identify as Terminating and Non-Terminating decimal.

Rational Number	In Decimal Form	Terminating	Non-Terminating (recurring)
$\frac{4}{3}$	1.333...		Non-Terminating
$\frac{3}{4}$	0.444...		
$\frac{5}{4}$	1.25	Terminating	

$\frac{25}{7}$	3.5714285...		
$\frac{3}{11}$	0.2727...		

EXERCISE 3.2

Q1. Convert into decimals and identify terminating or non-terminating decimals.

i. $\frac{7}{3}$ **ii.** $\frac{11}{6}$ **iii.** $-\frac{13}{9}$ **iv.** $\frac{15}{7}$ **v.** $\frac{14}{9}$

Q2. Convert into decimals and identify which is recurring and which is not.

i. $\frac{19}{20}$ **ii.** $\frac{17}{6}$ **iii.** $\frac{15}{8}$ **iv.** $\frac{25}{10}$ **v.** $\frac{2}{11}$

ROUNDING OFF NUMBERS

Rounding off is the process of simplifying a number while keeping its value **close to the original.** This is done to make numbers easier to use, especially when we don't need exact precision.

Why Do We Round Off?

We round off numbers to make them:

- Easier to work with in calculations.
- Simpler to understand and remember.

Example:
If the number is 47, and we round it off to the nearest 10, it becomes 50 because 47 is closer to 50 than to 40.

How to Round Off:

1. Look at the digit to the right of the place value you are rounding to.
2. If the digit is 5 or more, round up.
3. If the digit is less than 5, round down.

Example for Rounding to the Nearest Ten:
- 53 rounds to 50.
- 57 rounds to 60.

Rounding Decimals

1. Round 3.276 to the nearest tenth:
- Look at the digit in the **hundredths place** (7).
- Since 7 ≥ 5, increase the **tenths digit** (2) by **1**.
- Remove all digits after the **tenths place**.
- **Answer: 3.3**

2. Round 9.845 to the nearest hundredth:
- Look at the digit in the **thousandths place** (5).
- Since **5 ≥ 5**, increase the **hundredths digit** (4) by 1.
- Remove all digits after the **hundredths place**.
- **Answer: 9.85**

Key Points for Rounding Decimals:

1. Identify the place value you are rounding to (tenths, hundredths, etc.).
2. Look at the next digit to the right:
  - If the digit is **5 or greater**, increase the target digit by 1.
  - If the digit is **less than 5**, leave the target digit as it is.

3. Drop all digits to the right of the target place value.

EXERCISE 3.3

i. Round 754 to the nearest ten.
ii. Round 6,492 to the nearest hundred.
iii. Round 8.763 to the nearest tenth.
iv. Round 14.958 to the nearest whole number.
v. Round 0.456 to the nearest hundredth.

Complete the chart.

S. No.	Number	Up to two decimal places	Up to three decimal places	Up to four decimal places
1	0.05753	0.06	0.058	0.0575
2	2.421875			
3	0.83333			
4	1.76794			
5	0.9916667			
6	0.96257			
7	1.61538			
8	8.08547			
9	6.98677			
10	15.47648			